Systems Architecture

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http://blackboard.bournemouth.ac.uk/

http://dec.bournemouth.ac.uk/support/comp/sysarch
http://dec.bournemouth.ac.uk/support/sem/sysarch

Books: Main Text

- Peter Knaggs (2004)
  "*Systems Architecture: Lecture Slides*,"
  Bournemouth University.
- Stephen Welsh (2004)
  "*Systems Architecture: Term 1 Lab Book*,"
  Bournemouth University.
- Peter Knaggs and Stephen Welsh (2004)
  "*ARM Assembly Language Programming*,"
  Bournemouth University.
- Alan Clements (2000)
  "*The Principles of Computer Hardware*," (3/e)
  Oxford University Press, 0-19-856453-8.

Books: Supplementary Texts

- Robert Newman, *et. al.* (2002),
  "*Computer Systems Architecture*,"
  Crucial, 1-903337-07-0.
- D C Green (1999),
  "*Digital Electronics*,"
  Longman, 0-582-31736-3.
- Alex van Someren and Carol Atack (1994),
  "*The ARM RISC Chip: A Programmer's Guide*,"
  Addison-Wesley, 0-201-62410-9.

Books: Alternative Texts

- B S Chalk, Antony Carter and Robert Hind (2003), "*Computer Organisation and Architecture: An Introduction*," (2/e), Macmillan Press, 1-403-90164-3.
- Mark Burrell (2003), "*Fundamentals of Computer Architecture*," Palgrave, 0-333-99866-9.
- Andrew S Tanenbaum (1999), "*Structured Computer Organization*," (4/e) Prentice Hall, 0-13-020435-8.
- William Stallings (2003), "*Computer Organization and Architecture*," (6/e) Prentice Hall, 0-13-049307-4.

Autumn Term

Week	Topic	Chapter	Lab Topic
1	Introduction	1/2	Intro to Digital Works
2	Basic Logic Gates	3	Combinational Logic
3	Karnaugh Maps	4	Karnaugh Maps
4	Sequential Logic / Flip-Flops	5	Sequential Logic
5	Memory / Registers	6	Useful Circuits
6	The Tri-State / Bus	7	Encoders / Decoders
7	Integers	8	Binary Arithmetic
8	Characters / Strings	–	Character Strings
9	Decimal Numbers	9	Floating Point
10	The Central Processor Unit	1/2	Assignment

Spring Term

Week	Topic	Chapter	Lab Topic
11	The ARM Processor	3/4	Intro to ARMulator
12	Addressing Modes	5	Addressing Modes
13	Data Movement *Assignment One Hand-In*	7	Assembler Programs
14	Logic	8	Logic Programs
15	Structure / Loops	9	Looping Programs
16	*Project Week*	–	Assignment
17	Strings	10	Assignment
18	The Stack	14	Assignment
19	Subroutines *Assignment Two Hand-In*	15	The Stack
20	Interrupts / Exceptions	16	Subroutines
21	CISC / RISC	–	Exceptions

Week	Topic	Chapter	Lab Topic
22	Input / Output Methods	–	I/O Methods
23	Input/Output Devices	–	I/O Devices
24	Revision	–	Review of Past Papers
25	Exam Technique	–	Review of Past Papers